News and Articles

SUBMARINERS HELPING SUBMARINERS



The Submariner Tradition

Why Submariners Advocacy Group

The Unseen Burden - Amendment 2: Physiological Effects

The Unseen Burden - Amendment 1: Physiological Effects

The Unseen Burden: Toxic Exposures and Health Impacts

Submarine Toxic, Hazardous, and Radiation Exposures

(NEW News/Article)

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Home

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Navigation

About
Contact
Donate
Home
News and Articles
Online Store

About

Members of the Silent Service have been faithful to their oaths and training and have been silent for far too long. SAG will advocate for all Submariners.

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The Submarine Tradition



There is something about the submarine service. It tends to create a bond between those who have served that is born of trust. Every Submariner who ever put to sea and submerged has placed in the hands of another, their very lives. The bond is not one of close friendship, even though those do grow out of the time spent aboard a boat. It is one of mutual respect – blind to color, ethnicity, religion, nationality and gender. The bond is one of personal responsibility. Not everyone has it. Some don't even know what personal responsibility is. Submariners know what it is. It is their way of life. The trust and respect and sense of personal responsibility is… The Submarine Tradition.

Home

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Navigation

About
Contact
Donate
Home
News and Articles
Online Store

About

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Why Submariners Advocacy Group?



We are an association of veteran Submariners, all having earned and received the US Navy designation of “Qualified in Submarines” (SS). All our Board Members, Founders, future Advocates, and first contact associates have been there and done that on US Navy submarines, wherever “there” was and whatever “that” may have been. We have undertaken the arduous Submarine Qualification Program, been tested and tried by our shipmates, and found to be worthy of the moniker “Submariner.”

All advocates of the Submariners’ Advocacy Group (SAG) are committed to being a voice for all of our Brother and Sister US Navy Submariners. Submariners share a very unique bond forged in the depths of the oceans. We cannot adequately describe the experience to those who have never experienced life inside a submarine. Often, some aspects, knowledge, and specifics can never be shared with anyone outside of those who have served in the Silent Service. There is no membership application to fill out. You are eligible to be a part of the SAG because of the Dolphins you wear on your chest, you are “Qualified in Submarines,” you are a Submariner. If you are a Submariner you are automatically included.

SAG will be engaging the services of our future Affiliated Healthcare Professional Partners. These partners will be specialists in diving deeper into health issues that may be associated with our duties onboard US Navy submarines, specifically, Pulmonologists, Cardiologists, Oncologists, Dermatologists, Toxicologists, Ear-Nose-Throat specialists, Neurologists, and more. While these partners may not have served on submarines, SAG will provide them with all the information necessary to assist them in assisting Submariners.

SAG will seek Affiliated Research Partners to analyze the nationwide Submariner Registry and the data collected from US Submariners throughout the nation or wherever they may live. These Research Partners will be institutions with outstanding reputations and world-class research facilities. Further, SAG will be seeking partnerships with Veteran Service Organizations to leverage their services to assist veterans. This will include lobbying to get legislation passed addressing proper VA services, recognition of submarine-connected disabilities, and additional research conducted to better care for and protect all Submariners, both veterans and active duty.

These sound like lofty, “pie in the sky” goals, and they are. Yes, this won’t happen overnight and will require concerted efforts from all of the SAG team, but nothing will stop SAG from becoming the strongest voice on behalf of our Brothers and Sisters who voluntarily served in the US Submarine Force. We are the Elite of the Fleet and deserve elite recognition and services.

Home

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Navigation

About
Contact
Donate
Home
News and Articles
Online Store

About

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SUBMARINE TOXIC, HAZARDOUS, AND RADIATION EXPOSURES



Impact of toxic exposure on crews

Submarine crews face unique challenges due to potential toxic exposures from various operational environments. These exposures can lead to serious health issues, impacting both physical and mental well-being. Understanding these risks is crucial for advocacy and ensuring a safer environment for current and future submariners. Addressing these concerns can help promote awareness and drive necessary changes within the submarine community and veterans.

U.S. Navy submarines are the most technologically advanced and stealthy warship in the world. For 125 years, Submariners have confidently and proudly volunteered to sail under all the world's oceans to protect our national interests, prevent nuclear war and provide a silent and unseen global presence.

A small group of veteran Submariners began discussing the various toxins and hazardous chemicals they were exposed to while deployed onboard submerged submarines.

Eventually, these discussions turned to the countless stories of brother and sister Submariners who had developed health issues during or after their Naval service.

These discussions led to the formation of the Submariners' Advocacy Group (SAG) in April 2024.

SAG was founded by 15 Submariners who recognized the need for action on behalf of the estimated <300,000 Submariners in the U.S. population.

Submariners

Every Submariner is a volunteer. Not only are they expected to become an expert in their career field, but they must also become an expert in the submarine itself.

Every member of the crew must learn the purpose, function, and how every system on board works. They must understand the basics of nuclear power, electrical distribution, ventilation and atmosphere control, hydraulics, weapons systems and how the boat itself is constructed.

They must become experts in combatting casualties like flooding and fires, as well as know where every piece of damage control equipment on board is located and most importantly, how to use it. In addition to those tangible skills, they must form a mutual bond of trust and respect. This applies to every member of the crew from the Commanding Officer to the newly reported junior Sailor.

Every Submariner must undergo an intense and challenging qualification process designed to test them mentally and physically. Once they have demonstrated their knowledge, trust, and gained the respect of the crew, they earn the coveted warfare insignia of a U.S. Submariner – Dolphins.

Earning one's Dolphins demonstrates a commitment to excellence unmatched by any other branch in the U.S. Military

Submarine Atmosphere and Monitoring

In 1955, nuclear power changed submarine technology forever. With an unlimited source of power for propulsion and electricity, nuclear power gave submarines the ability to remain submerged for much longer periods of time.

Maintaining a habitable atmosphere for the crew was a critical factor for both supplying the oxygen required to breath and to remove byproducts such as carbon dioxide. To prolong the length of time a submarine could stay submerged, three critical pieces of equipment were developed: The Oxygen (O2) Generator, Carbon Dioxide (CO2) Scrubber, and the Carbon Monoxide – Hydrogen (CO-H2) Burner.

Oxygen is created by the O2 Generators by using the process of electrolysis from seawater that has been distilled, filtered, and deionized. It is stored in large storage tanks vented directly to the ship's atmosphere.
Carbon dioxide is removed from the atmosphere by utilization of the chemical monoethanolamine (MEA or amine). At room temperature MEA absorbs CO2. Heating MEA allows it to release CO2 efficiently before disposal through a diffuser overboard.

The CO-H2 Burner uses a very high temperature baseplate that converts carbon monoxide into carbon dioxide and hydrogen gas into water vapor. These critical pieces of atmosphere control equipment work together with the ship's ventilation system.

The ship's ventilation system consists of large recirculation fans, dampers, blowers, as well as the ship's emergency diesel engine. The ventilation system can be used to evacuate smoke and toxins resulting from casualties by using the diesel engine as a high-powered exhaust fan. Dampers and valves can be configured to force air from low pressure blowers into the submarine's ballast tanks while surfacing or running on the surface.

A sophisticated monitoring system known as CAMS keeps the submarine's atmosphere in check. Sensors in every habitable space onboard sample the atmosphere and monitor for abnormalities. Atmosphere readings are reviewed every hour by a senior member of the watch team. In the event an out of specification reading is detected, an alarm will sound, and immediate actions are taken by the crew to regain control of the ship's atmosphere.

Based on publicly available information, it can be presumed that atmospheres on U.S. Navy submarines have not been adequately studied, measured, or monitored. There is a need for the collection of more substantial data on the long-term effects of exposure to the submarine atmosphere and the contaminants it contains.

Toxic & Hazardous Chemicals

There are potentially 130 or more chemicals that could be classified as submarine atmospheric contaminants. In 2007, an NRC report only focused on 26 individual chemicals or gases, only 20% of the known or possible contaminants present in the submarine atmosphere.

Surveys recommended by 1983 and 1988 reports were not performed by the time the NRC COT subcommittee was established in 2002. Failure to conduct these studies has resulted in an inadequate understanding of the toxicity of the atmospheres onboard submarines. As a result of the studies having not been conducted, tens of thousands of Submariners have been continuously exposed to these hazardous chemicals.

The three-volume report of 2007 to 2009 admitted the failure to address exposure to chemical mixtures and the potential for antagonistic, additive, or synergistic interactions between contaminants present in the submarine environment.

The findings and recommendations were based on widely accepted industry standards regarding exposure limits. However, they failed to take into consideration that onboard a submerged submarine, human exposures are continuous for 24 hours a day for months on end in the sealed environment of the submarine.

This report focuses on only a few of the known gases, chemicals, and other atmospheric contaminants present in U.S. Navy submarine atmospheres. The following gasses and chemicals will be the focus of the SAG Report:

Oxygen

Submariners are routinely subjected to O2 levels below the normal atmospheric level of 20.9% and often in hypoxic, or O2 deficient levels less than 19.5%

Carbon Dioxide

The atmosphere on a submerged submarine often runs with higher than accepted exposure limits for CO2. The NRC recommended 90-day levels be increased to 0.8%, 20 times greater than the amount of CO2 contained in normal air (0.04%).

Monoethanolamine (MEA / Amine)

Amine vapor escapes from the CO2 Scrubbers and permeates every porous material onboard the submarine; skin, clothing, bedding, and paper. There have been no studies on the long-term exposure to amine experienced by Submariners. Amine is acutely toxic.

2,6-di-tert-butyl-4-nitrophenol (DBNP)

2,6-di-tert-butyl-4-phenol (DBP) was used as a preservative in 2190 TEP, the submarines' main lube oil. When TEP was released into the submarine atmosphere, went through electrostatic precipitators, DBP was converted to DBNP which has been identified as an acute toxin.

Benzene

Benzene has been widely studied and is identified as a carcinogen and is acutely toxic. It is used in many applications throughout submarines.

Asbestos

The hazards associated with asbestos have become common knowledge. In the past it was used extensively in insulation and other applications as a fire retardant on submarines.

Home

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Navigation

About
Contact
Donate
Home
News and Articles
Online Store

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The Unseen Burden: Toxic Exposures and Health Impacts

Watch the Presentation



Press Release
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Read the official announcement of:

The Unseen Burden report release

Abstract
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Quick overview of the report's key findings and recommmendations

Full Report
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The complete 59-page comprehensive research report (The Unseen Burden)

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Home

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Navigation

About
Contact
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Home
News and Articles
Online Store

About

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The Unseen Burden - Amendment 1: Physiological Effects

Amendment 1: Physiological Effects

Following the release of our landmark report The Unseen Burden, this amendment provides critical analysis of the physiological impact of toxic exposures on U.S. Navy Submariners.

Amendment 1 - Physiological Impact Analysis

Amendment 1
Download PDF

Comprehensive analysis of physiological impacts from toxic exposures on submariners

Key Focus Areas

This amendment expands on the original report by providing:

- Detailed physiological analysis of how toxic exposures affect submariner health
- Medical evidence linking submarine atmospheric conditions to specific health outcomes
- Scientific documentation supporting the need for expanded healthcare coverage
- Recommendations for improved health monitoring and care protocols

Related Documents

For complete context, review the original report and supporting materials:

Abstract
Download PDF

Quick overview of the report's key findings and recommmendations

Full Report
Download PDF

The complete 59-page comprehensive research report (The Unseen Burden)

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Navigation

About
Contact
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Home
News and Articles
Online Store

About

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The Unseen Burden - Amendment 2: Compounded Stressors



Amendment 2: Compounded Stressors

Following the release of our landmark report The Unseen Burden - Amendment 1, this amendment provides an analysis of how compounded stressors and other issues are associated with the toxic, hazardous, and radiation exposures.

Amendment 2 - Compounded Stressors and the Physiology of “Human Error”

Amendment 2
Download PDF

Comprehensive analysis of how multi-stressor environment and toxic exposure are a cumulative risk.

Key Focus Areas

This amendment expands on the original report and amendment 1 by providing:

- Effects of cumulative risk exceeding human biological thresholds
- Disruption of Circadian Rhythms and the Biological, Psychological, and Cognitive impacts
- “Synergistic Trap”, how atmospheric conditions interact and magnify the impairment of a sleep-deprived brain
- Matrix of Multi-Stressor Interactions

Related Documents

For complete context, review the original report and supporting materials:

Abstract
Download PDF

Quick overview of the report’s key findings and recommmendations

Full Report
Download PDF

The complete 59-page comprehensive research report (The Unseen Burden)

Amendment 1
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The Unseen Burden - Amendment 1: Physiological Effects

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Home

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Navigation

About
Contact
Donate
Home
News and Articles
Online Store

About

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Donation Program

SUBMARINERS HELPING SUBMARINERS

Support Our Submariners

Why do we ask for donations? Well, quite frankly, it's a bit expensive to meet with key members of important organzations to introduce them to the needs of Submariners.

Transparancy

What exactly are we spending your donations on?

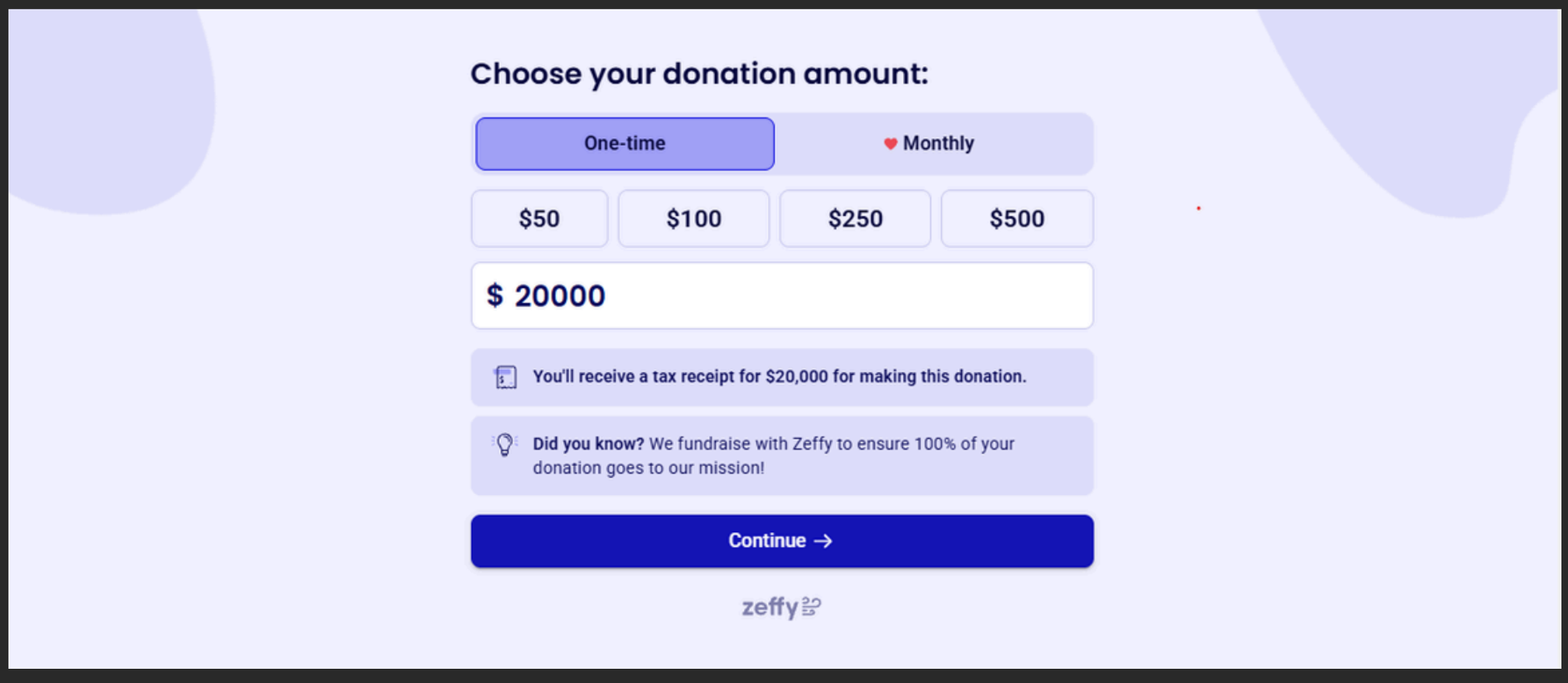
- Travel: Providing resources and assistance to families, ensuring they have the support they need while their loved ones are away.
- Business Operations: Funding educational programs and training to enhance the skills and career opportunities of submariners.

Real Stories, Real Impact

Do you have a story of how the Submariners' Advocacy Group has helped you or someone you know?
Share it with us at public-affairs@subadvocates.org

How to Donate

We have partnered with Zeffy to make donating easy and secure. Use the form below to donate to the Submariners' Advocacy Group.



Other Ways to Help

- **Volunteer**: Join our team and help us support submariners and their families.
  - **Researchers**: Help us gather information and data to support our mission.
  - **Report Writers**: Help us create reports and white papers to share with key members of Congress and other organizations.
  - **Technology Expertise**: Website development, social media management, web application development, and more…
  - **Public Relations**: Help us spread the word about our mission and the needs of submariners.
- **Advocacy**: Help us meet with key members of Congress and other organizations to advocate for submariners.
- **Spread the Word**: Share our mission with your friends and family to increase awareness and support.

Thank you for your support!

Home

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Navigation

About
Contact
Donate
Home
News and Articles
Online Store

About

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Navigation

About
Contact
Donate
Home
News and Articles
Online Store

About

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